

QUIZ 1

1. Which hashing algorithm is used in Bitcoin for Proof of Work? **(1 mark)**

a) MD5 b) SHA-1 c) SHA-256 d) AES

2. The Merkle root is in the: **(1 mark)**

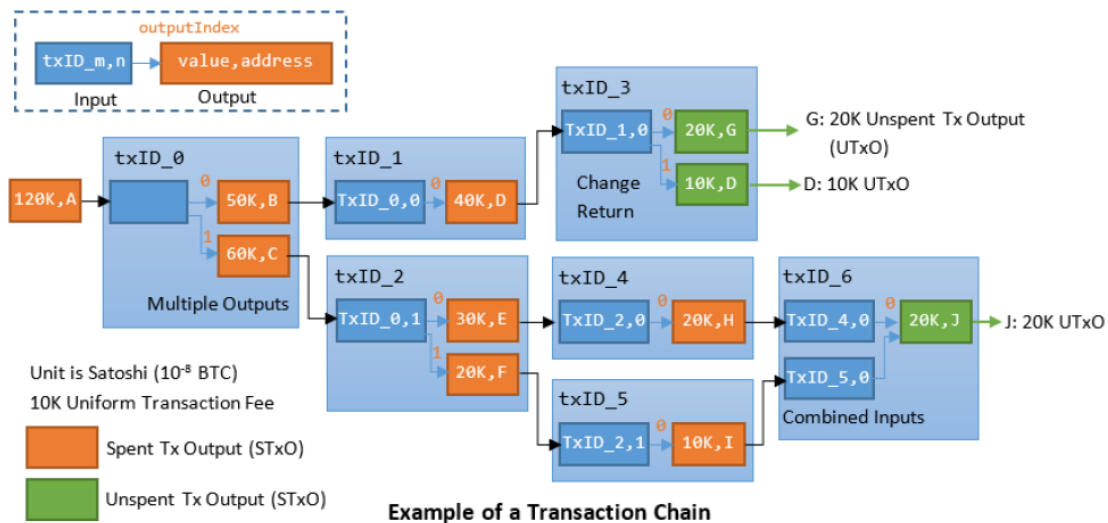
a) Transaction list b) Wallet database c) Block header d) Mining pool

3. Digital signatures provide: **(1 mark)**

a) Confidentiality only b) Integrity and authentication c) Data compression d) Hashing

4. How can we differentiate between hashing and encryption in blockchain technology. **(2 marks)**

5. Consider the following diagram of bitcoin transaction chain: **(5 marks)**



- In txID_0, how many outputs are created?
a) 1 b) 2 c) 3 d) 4
- Trace the transaction path from 120K, A to 20K, J. List the transaction IDs involved.
- If txID_6 did not combine inputs, would it be able to generate 20K, J? Justify your answer.